Math 133: Quiz 3

Friday, September 25, 2026 25 minutes 10 points

Name: _____

Show your calculations and include units where appropriate. A simple scientific calculator is allowed.

1. (2 points) The table gives a workshop's total cost $C(q)$, in dollars, for producing q items.

q (items)	9	10	11	12	13	14	15
$C(q)$ (dollars)	95	100	106	112	119	127	135

   (a) (1 point) Find the marginal cost of the 11th item (the exact additional cost of producing that item).

   Marginal cost: _____

   (b) (1 point) Find the average rate of change of cost from 10 to 15 items. Include units.

   Average change of cost _____

2. (3 points) A stock starts at \$60 and grows by 4% each year.

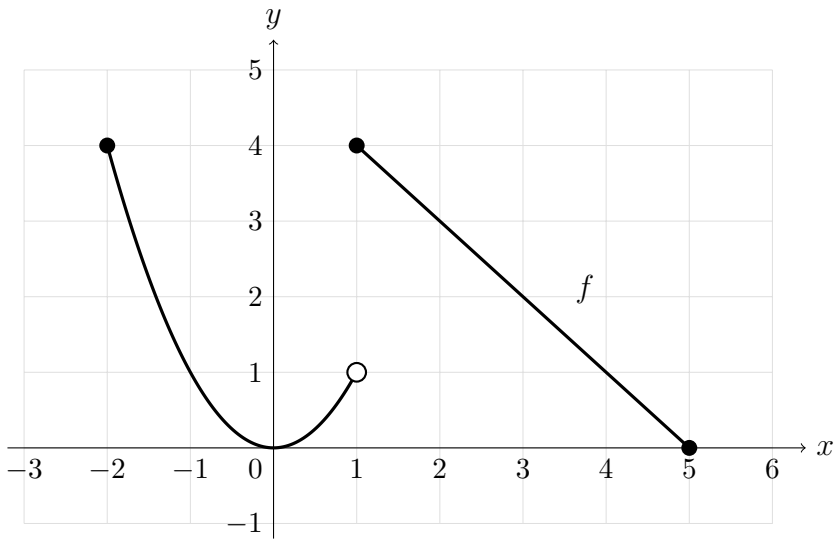
   (a) (1 point) Using an exponential model, write a function $V(t)$ that represents the value of the stock at time t, measured in years.

   $V(t) =$ _____

   (b) (2 point) Find the number of years t it takes to for the stock to grow to \$100. Your answer should be exact, if you use logarithms do not evaluate them.

   $t =$ _____

3. (5 points) Use the graph of f below for all parts. Open circles are excluded; filled circles are included.



(a) (2 points) Find the slope and an equation of the secant line through $(-2, f(-2))$ and $(0, f(0))$.

(b) (2 points) On the interval $[0, 4]$, find both the net change in f and the average rate of change of f.

Net change: _____ Average rate of change: _____

(c) (1 points) Evaluate each limit from the graph. If a limit does not exist, write DNE. Briefly explain any DNE answer.

$\lim_{x \to -1} f(x) =$ _____ $\lim_{x \to 1^+} f(x) =$ _____

$\lim_{x \to 1^-} f(x) =$ _____ $\lim_{x \to 1} f(x) =$ _____